Dynamic Category Preference Learning: Tackling Long-Tailed Semi-Supervised Segmentation in Remote Sensing

Junjun Li, Yong Liu, Shiheng Zhang, Baoqi Yu, Zijun Zhou, and Jinze Liu

***Abstract*—Remote sensing (RS) image segmentation faces persistent challenges due to limited labeled data and the presence of a long-tailed distribution. While semi-supervised learning (SSL) can leverage unlabeled data, it often struggles to address the severe class imbalance that manifests as a long-tailed distribution. To tackle this, we propose the dynamic category preference-aware semantic segmentation framework (CPSeg). Unlike existing methods that estimate a model's category preference on pseudo-labels or rely on prior information, CPSeg introduces a simple yet effective approach to directly model this preference. By feeding a pattern-free image (PFI) into the model, we capture its current tendencies toward certain categories without any prior knowledge. This dynamically learned category preference drives two key components of CPSeg. First, it enables adaptive weight allocation, which adjusts the training focus by assigning higher weights to low-frequency categories, promoting balanced learning across all categories. Second, CPSeg introduces a Category-Wise Pairwise Memory Bank (CPMB) and integrates category preference information to perform dynamic preference-aware data augmentation, making the augmentation process more suitable for addressing the long-tailed distribution problem. When compared with other advanced approaches, CPSeg exhibits superior performance, achieving mean intersection over union (mIoU) scores of 74.50% on the GID-15 and 59.26% on the MSL dataset, respectively, under a challenging 1/8 labeled data setting, highlighting its strong potential for addressing long-tailed distribution and class imbalance in RS segmentation.**

***Index Terms*—Adaptive perturbation augmentation, category preference-aware, remote sensing (RS) images, semi-supervised semantic segmentation.**

I. Introduction

REMOTE sensing (RS) technology captures high-resolution imagery using satellite or UAV-mounted sensors, offering rich geospatial data [1], [2]. The rapid development of computer vision has advanced RS image analysis significantly. Deep learning methods are widely used in change detection, object detection, classification, and semantic segmentation. Semantic segmentation classifies each pixel into specific semantic regions, such as buildings, roads, vegetation, or water. It is vital for urban planning, 3-D city modeling, land surveys, disaster warning, and traffic coordination [3], [4], [5].

Deep networks greatly enhance RS segmentation, especially for high-resolution and complex scenes. CNNs and their variants like FCN [6], U-Net [7], and DeepLab [8], [9], [10], [11] have achieved impressive results. However, they rely heavily on pixel-level annotations, which are costly and labor-intensive [12]. Annotating RS data requires expert knowledge, further increasing difficulty. In contrast, large amounts of unlabeled data are easy to obtain. Semi-supervised semantic segmentation offers a way to train models with limited labeled data and abundant unlabeled data, reducing annotation dependence.

Current semi-supervised methods mainly follow two strategies: consistency regularization (CR) [13] and pseudo-labeling (PL) [14]. CR relies on the clustering assumption [15], enforcing prediction stability under perturbations, such as input [16], [17], [18], feature [19], or network noise [20], [21]. PL uses a model trained on labeled data to generate pseudo-labels for unlabeled data, which are then used in self-training [14], [22], [23], [24]. FixMatch [25] combines CR and PL, using confident pseudo-labels from weak augmentations to guide strong augmentations. Its fixed threshold strategy brings substantial improvements on various tasks. Many follow-up studies further develop this framework from different perspectives [26], [27], [28].

In RS segmentation, many works adopt this semi-supervised learning (SSL) paradigm to better exploit unlabeled data [29], [30], [31], [32], [33]. For example, WSCL [31] proposes sparse dual-view CutMix-based augmentations to enrich representations and improve pseudo-labels. CrossMatch [33] combines cross-view weak-to-strong consistency at both image and feature levels, integrating diverse cues to enhance generalization and segmentation quality. These methods bring new ideas to semi-supervised RS segmentation.

Although significant improvements have been achieved, a persistent challenge in real-world RS image segmentation is the severe class imbalance, which characteristically manifests as a long-tailed distribution. This is exemplified in the GID-15 dataset (Fig. 1), where under a 1/8 labeled data setting, "head" classes like irrigated land are over 120 times more frequent than "tail" classes like shrub land (SL). This long-tailed nature critically impairs the ability of deep learning models to learn discriminative features, as they require sufficient examples for

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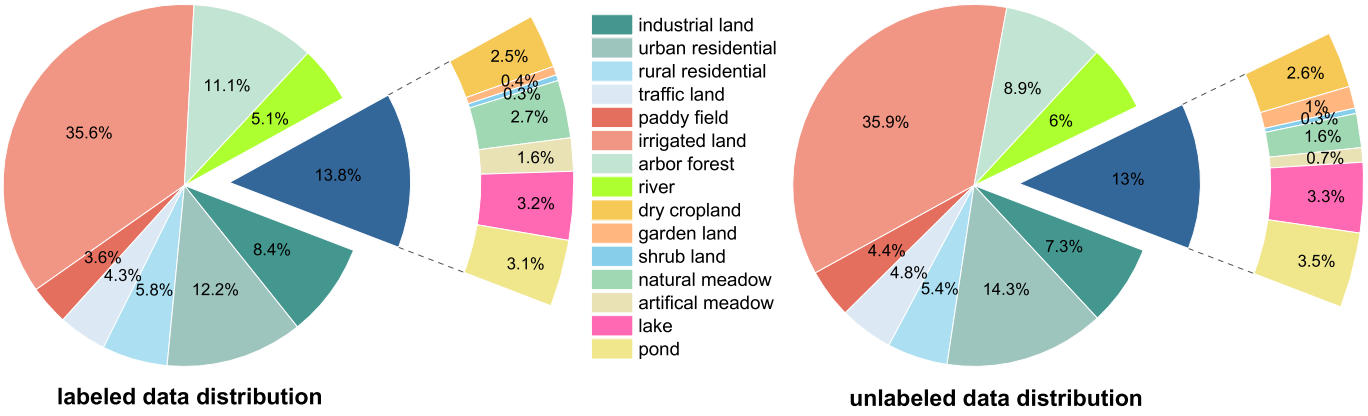


Fig. 1. Class distribution of (left) labeled and (right) unlabeled training data in GID-15 under a 1/8 data ratio, showing the imbalance among different categories.

each class to build robust representations. During optimization, the overall loss function becomes dominated by the high-frequency head classes. Consequently, the model's gradients are biased, compelling it to prioritize and overfit to these majority classes. Conversely, the supervisory signals from tail classes become faint and are largely ignored, leading to poor feature representation for rare categories and severely degrading the model's overall performance and generalization ability.

The long-tailed distribution is a foundational challenge common to both fully-supervised (FSL) and SSL. In either setting, the training process is invariably dominated by head classes, leading to poor generalization on tail classes. However, the problem becomes critically exacerbated in the SSL paradigm. This is because the core mechanism of SSL, which involves generating pseudo-labels for unlabeled data, introduces a pernicious feedback loop when combined with data imbalance. A model initially biased toward head classes generates more frequent and confident pseudo-labels for these same classes, creating a vicious cycle of bias amplification where the initial imbalance is actively reinforced. Consequently, tail classes suffer from a dual scarcity of supervision: their ground-truth labels are rare, and reliable supervisory signals from unlabeled data are also severely diminished. Furthermore, conventional long-tailed strategies from FSL, such as reweighting, often require an accurate estimation of the true class distribution. This is infeasible in SSL, as statistics from a small, unrepresentative labeled set are inherently biased, and those derived from pseudo-labels are already skewed by the model's own flawed predictions.

To address these specific challenges inherent to the semi-supervised context, we propose CPSeg, a semantic segmentation framework that incorporates category preference modeling into both weight allocation and perturbation augmentation. Unlike methods that depend on skewed data statistics, CPSeg is designed to directly diagnose the model's intrinsic biases in real-time. This dynamic preference drives two core modules: adaptive weight allocation based on category preference (AWACP) and adaptive perturbation augmentation based on category preference (APACP). Category preference is measured using a pattern-free image (PFI) to reflect the model's current bias. AWACP leverages this signal to adjust loss weights, suppressing overfitting to head classes while enhancing learning for tail classes. Concurrently, APACP uses this preference information and a Category-Wise Pairwise Memory Bank (CPMB) to guide a dynamic data augmentation process, promoting learning on tail categories and mitigating the amplified bias inherent in long-tailed SSL.

The main contributions of this article are as follows.

1) We introduce a novel dynamic category preference modeling approach that diagnoses the model's intrinsic biases in real-time by analyzing its response to a PFI. This offers a more accurate and dynamic signal for tackling the long-tailed problem compared to methods relying on static data statistics.
2) We propose an AWACP strategy. This module leverages the real-time category preference to dynamically adjust loss weights, effectively suppressing overfitting to head classes and enhancing learning for tail classes.
3) We design an APACP strategy. This module uses the category preference signal to guide a dynamic data augmentation process, selectively sampling and pasting tail-class examples to enrich the training data and improve their feature representations.

II. Related Work

A. Semi-Supervised Semantic Segmentation

Semi-supervised semantic segmentation aims to reduce annotation costs by leveraging abundant unlabeled data alongside a small labeled set. A primary strategy is self-training, which uses high-confidence pseudo-labels from model predictions to enrich the training set. To improve pseudo-label quality, techniques, such as dual-decoder models [34], refinement modules [35], and stability-based selection mechanisms [14] have been proposed. Another established paradigm involves generative adversarial networks (GANs), where a segmentation network acts as a generator and a discriminator works to distinguish generated segmentation maps from ground-truth annotations, thereby enforcing structural plausibility [36], [37]. However, the dominant approach is CR, which enforces prediction stability under perturbations. This

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is typically achieved through a weak-to-strong augmentation framework, pioneered by FixMatch [25], where confident pseudo-labels from weakly augmented images supervise predictions on strongly augmented counterparts. This principle has been advanced by frameworks like CPS [21], which uses a co-training paradigm for mutual supervision, UniMatch [27], which streamlines the architecture, and Allspark [38], which introduces Transformer-based designs and semantic consistency. Other notable techniques include feature perturbation [19], image-label mixing [17], [18], and managing unreliable predictions [39].

These methodologies have been extensively adapted to address the unique challenges of RS image segmentation. In the self-training direction, LSST [40] introduces a linear sampling-based pseudo-label selection mechanism, while DAST [41] integrates dynamic sampling and distribution matching. For CR, WSCL [31] uses a sparse dual-view strategy to extend training samples, and MCSS [32] applies multiperturbation constraints to enhance robustness in complex scenes. To better handle the intricate characteristics of RS data, such as highly similar categories and scale variation, more specialized methods have emerged. For example, CrossMatch [33] introduces a cross-view consistency mechanism to resolve ambiguity. Xin et al. [42] proposed a dual-teacher structure combining CR with bias contrastive learning, while MUCA [43] incorporates multiscale uncertainty constraints and a cross-teacher–student attention mechanism. These adaptations demonstrate a clear trend toward tailoring general semi-supervised techniques to the specific data properties inherent in RS. This trend continues in recent works; for instance, HARC [44] proposes a hierarchical augmentation strategy specific to RS data alongside region-aware contrastive learning to enhance robustness against noisy pseudo-labels. Another emerging direction is the incorporation of external prior knowledge, exemplified by PGNet [45], which leverages the segment anything model (SAM) to guide feature learning and provide explicit edge supervision, thereby improving the segmentation of object boundaries.

B. Long-Tailed SSL

The challenge of long-tailed distributions in SSL, where a few "head" classes dominate the dataset, has spurred various strategies, primarily pioneered in image classification. The core ideas focus on correcting the pseudo-label bias induced by class imbalance. For instance, DARP [46] proposes to align the pseudo-label distribution with an estimated true data distribution via a refinery module. Based on the observation that tail classes often exhibit high prediction precision, CReST [47] introduces class-aware confidence thresholds, allowing more pseudo-labels to be selected for rare classes. Other approaches work on rebalancing the loss function, such as DRw [48], which adapts sample weights based on the temporal dynamics of pseudo-label counts, and TRAS [49], which uses a teacher–student framework where the teacher generates a more balanced pseudo-label distribution via instance-dependent logit adjustment to guide the student.

These foundational principles have been successfully extended and adapted to the more challenging task of semi-supervised semantic segmentation. For example, the DHC framework [50] leverages a co-training process between models with different initializations to generate diverse yet reliable pseudo-labels in 3-D medical imaging, guided dynamically by pseudo-label statistics. In the RS domain, AdaptMatch [51] constructs a foreground–background convergence difficulty-aware mechanism with an adaptive confidence threshold to account for varying learning difficulties across classes, thereby reducing pseudo-label bias. Similarly, methods like DAST [41] and DWL [52] integrate distribution matching and category-ranking-based weighting to specifically counter the dual challenges of pseudo-label noise and long-tailed distributions, aiming to alleviate learning bias by rebalancing the supervisory signals from both labeled and unlabeled data.

While these strategies have made significant strides, many prominent approaches, such as DHC [50], DAST [41], and AdaptMatch [51], operate on a similar principle. They typically depend on pseudo-labels generated from unlabeled data to indirectly infer the data distribution or the model's learning biases. This dependency can be a limitation, as pseudo-labels are often unreliable in the early training stages or under low-data regimes, potentially leading to noisy estimations. More critically, this indirect approach does not directly probe the model's own intrinsic and evolving category preference. This can lead to a mismatch between the applied remedial strategies and the model's true learning state, thereby impairing its ability to learn effectively from low-frequency categories. To bridge this crucial gap, our proposed CPSeg method introduces a novel mechanism that directly models and leverages this category preference.

III. Methodology

The proposed CPSeg is designed to address the long-tailed distribution issue in RS image segmentation tasks. CPSeg effectively mitigates the model bias caused by long-tailed distribution in both RS images and training data by employing intercategory adaptive weight distribution and category preference-based adaptive perturbation augmentation strategies. This approach enhances the model's learning capacity, particularly for low-frequency categories. The overall framework of the CPSeg method is shown in Fig. 2, which includes the following key components: first, the AWACP strategy dynamically adjusts category weights to suppress overfitting for high-frequency categories and strengthen learning for low-frequency categories; and second, the APACP strategy utilizes a CPMB to store and update original-labeled mask pairs (OLMPs) and adjusts sampling probabilities based on category preference, optimizing the data augmentation process.

A. Preliminaries

In the semi-supervised RS image semantic segmentation task, we have an unlabeled dataset $D^U = \{x_i^u\}$ and a labeled dataset $D^L = \{(x_i^l, y_i^l)\}$. Our loss function consists of two parts: supervised loss and unsupervised loss. In our proposed method, CPSeg, both the supervised and unsupervised losses share the same dynamic weight, which is computed based on category preference at the beginning of each epoch. The

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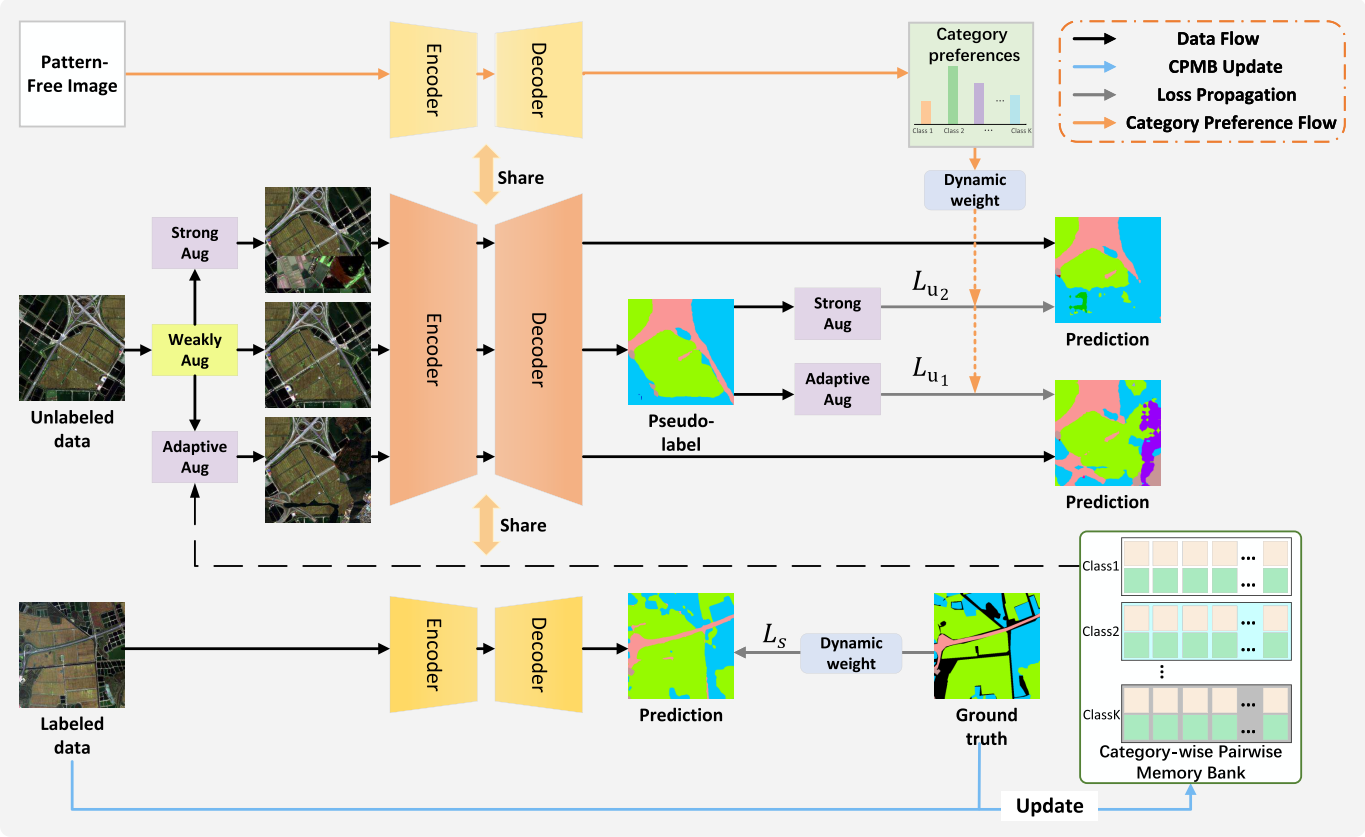


Fig. 2. Overview of the proposed CPSeg framework. CPSeg leverages a dual-branch for unlabeled data: a weakly augmented image is used to generate reliable pseudo-labels. These pseudo-labels then supervise two parallel and complementary branches. The first branch employs standard strong augmentation for general robustness. The second branch features our proposed adaptive perturbation augmentation, which is guided by category pReferences and a CPMB to alleviate the challenges of long-tailed distributions. For labeled data, supervision is provided via ground truth, which also updates the CPMB. The category preferences, derived from a PFI, guide the augmentation process and are used by our adaptive weight allocation strategy to determine dynamic loss weights.

specific computation is detailed in Section III-B. The overall loss function is defined as

$$L = L_s + L_u \tag{1}$$

where L_s represents the supervised loss on labeled data, and L_u denotes the CR loss on unlabeled data.

For labeled data, the supervised loss is formulated as

$$L_s = \frac{1}{N_l} \sum_{i=1}^{N_l} L_{\text{ce}} \left(F\left(x_i^l\right), y_i^l; \mathbf{w}_t \right) \tag{2}$$

where N_l is the number of labeled samples in the batch, x_i^l is a labeled image with its ground-truth annotation y_i^l, F is the segmentation model, and $\mathbf{w}_t$ is a vector of dynamic category weights at epoch t. The calculation of $\mathbf{w}_t$ is detailed in Section III-B.

For unlabeled data, we employ a CR loss guided by high-confidence pseudo-labels. First, for each unlabeled image x_i, we generate a weakly augmented version $x_i^{u_w}$ and its corresponding prediction map $y_i^{u_w} = F(x_i^{u_w})$. This prediction serves as the primary source for generating pseudo-labels. The model's prediction confidence is given by $c_i^w = \max(y_i^{u_w})$.

Our method implements this consistency principle through a dual-branch structure that supervises the model with complementary augmentations. The total unsupervised loss is the sum of two components, L_{u_1} and L_{u_2}, corresponding to an adaptive augmentation branch and a standard strong augmentation branch (using CutMix [17]), respectively,

$$L_u = \lambda L_{u_1} + (1 - \lambda) L_{u_2} \tag{3}$$

where λ is set to 0.5 in our experiments. Both loss components are computed using a shared formulation, but with different inputs. For each branch $j \in \{1, 2\}$, the loss is calculated as

$$L_{u_j} = \frac{1}{N_u} \sum_{i=1}^{N_u} I\left(c_i^w \geq \tau\right) \cdot L_{\text{ce}} \left(F\left(x_i^{s_j}\right), y_i^{s_j}; \mathbf{w}_t \right). \tag{4}$$

Here, $x_i^{s_j}$ is the augmented image for branch j. Crucially, its corresponding pseudo-label target, $y_i^{s_j}$, is derived by applying the same augmentation transformation used to create $x_i^{s_j}$ to the initial prediction map $y_i^{u_w}$. The indicator function $I(\cdot)$ applies this consistency training only to high-confidence samples. This dual-branch process encourages the model to produce consistent predictions across diverse and challenging augmentations, enhancing its robustness.

B. Adaptive Weight Allocation

In datasets characterized by a long-tailed distribution, the model is easily affected by the frequency of categories during training, resulting in high-frequency categories dominating the optimization process and insufficient learning of low-frequency categories. The conventional solution is to reduce

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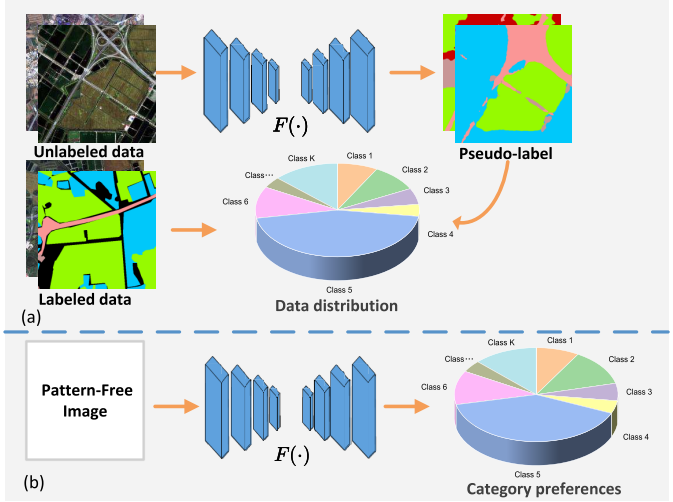


Fig. 3. Comparison of Data information acquisition methods. (a) Previous method for data distribution. (b) Our method for acquiring category preferences.

the model's dependence on high-frequency categories while enhancing the learning of low-frequency categories by allocating differentiated weights to different categories. The core of this lies in accurately estimating the true category distribution of the data. In general, high-frequency categories should be allocated lower weights to avoid dominating the optimization, while low-frequency categories need to be given higher weights to enhance learning. In a supervised learning framework, the statistical distribution of categories of the labeled data can be directly exploited for dynamic weight adjustment. However, in SSL scenarios, since the proportion of labeled data is much lower than that of unlabelled data, relying only on labeled data for distribution estimation may lead to serious bias, making it difficult to effectively reflect the real data distribution.

As shown in Fig. 3, previous studies have attempted to use PL to infer the data distribution of unlabelled data and make category weight adjustments based on it. Specifically, DHC [50] measures the category distribution by calculating the number of pseudo-labeled category voxels for unlabelled data and uses this information for dynamic weight allocation. However, since the generation of pseudo-labels relies on the current model's prediction, the pseudo-label distribution may be affected by the model's implicit category preference, which may introduce systematic bias and reduce the effectiveness of the weight adjustment.

To address this problem, we propose an adaptive weight allocation method based on category preference, which constructs a more stable estimate of the category distribution by detecting the model's own category preference, thereby reducing the influence of pseudo-label noise in the weight calculation process.

Inspired by CDMAD [53], we introduce a dynamic category preference-aware method based on PFI. Specifically, we construct an all-white image I_w (an image without structural information) and input it to the current model $F(\cdot)$ to obtain the original prediction logits Z^W

$$Z^W = F\left(I_w\right). \tag{5}$$

Subsequently, we apply the argmax operation on Z^W to obtain the final category prediction y^W, where y^W has the same spatial dimensions as the input image $[H, W]$, and each pixel value in y^W corresponds to the predicted category at that location. Next, we compute the number of pixels N_k corresponding to each category in y^W

$$N_k = \sum_{i=1}^{H}\sum_{j=1}^{W} I\left(y_{i,j}^W = k\right). \tag{6}$$

As shown in (6), where $I(\cdot)$ is the indicator function. Equation (6) computes the number of pixels classified as category k in the PFI by the model, and N_k reflects the model's category preference for category k.

After obtaining the category preferences of the model, we introduce a dynamic weight allocation strategy to suppress the overfitting of the model to high-frequency categories and to enhance the learning ability of low-frequency categories. At the beginning of training, the initial weights of all categories are set to the total number of categories K. The initial weights are allocated as

$$\tilde{w}_1^k = K \quad \forall k \in 1, 2, \ldots, K. \tag{7}$$

This initialization ensures that the weights of the categories are balanced in the early stages of training and avoids the model over-focusing on certain categories. Initial training instability due to category bias is avoided.

In each training epoch, we use PFI to estimate the model's category preferences and adjust the category weights based on its preference. Specifically, the frequency of occurrence of category k, N_k, is measured from the model's predictions under PFI, reflecting the extent of the model's inherent preference for that category. Based on this preference, we calculate the dynamic weights of the categories. Based on this preference, we calculate the raw weight for the kth category at the tth epoch as

$$w_t^k = \frac{1}{N_k + \epsilon} \tag{8}$$

where ϵ is a very small positive number (e.g., 1×10^{-7}) used to prevent division by zero errors. Equation (8) ensures that low-frequency categories receive large weights, while high-frequency categories have relatively low weights, thus balancing the category learning process.

To avoid significant fluctuations in weights, we use an exponential moving average (EMA) to smooth the update of weights, defining the final category weights for the kth category at the tth epoch as

$$\tilde{w}_t^k = \beta \tilde{w}_{t-1}^k + (1-\beta)\, w_t^k \tag{9}$$

where β is the momentum parameter, which is set to 0.99 in the experiment. The EMA mechanism can smooth the magnitude of category weight changes, mitigate instability caused by fluctuations in category preferences during training, and make the weight adjustment more robust.

Finally, the dynamic weight vector $\mathbf{w}_t$ used in the loss functions [see (2) and (4)] is constructed from these smoothed scalar weights for all K categories: $\mathbf{w}_t = [\tilde{w}_t^1, \tilde{w}_t^2, \ldots, \tilde{w}_t^K]$. This vector is then passed to the weighted cross-entropy loss to balance the training process.

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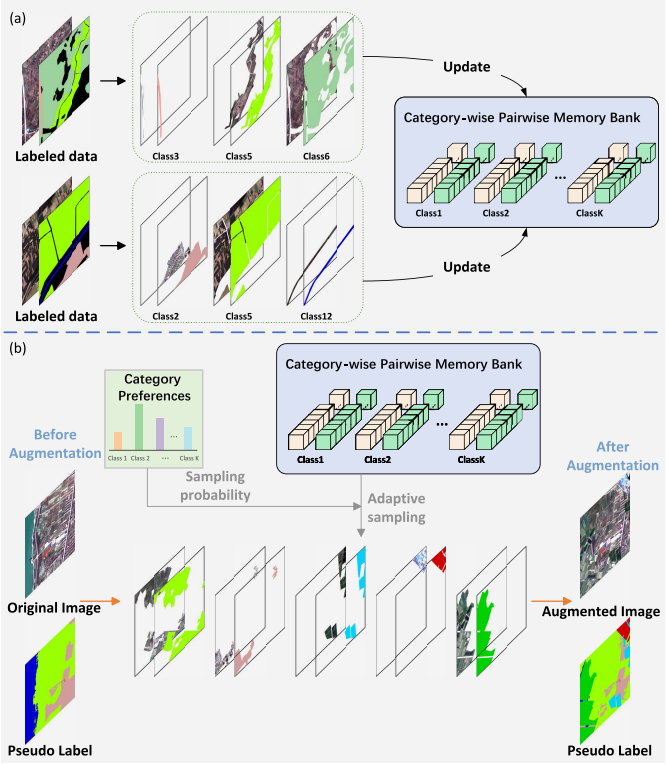


Fig. 4. Illustration of the CPMB and the APACP process. (a) Update mechanism of the CPMB, where OLMPs are extracted from labeled data and populated into category-specific queues. (b) APACP mechanism. Guided by dynamic category preferences, the CPMB provides OLMP for adaptive sampling to augment unlabeled data, thereby enhancing the representation of tail categories in the training data.

C. Adaptive Perturbation Augmentation

We first construct a CPMB to serve as a dynamic source of information for augmentation. As illustrated in Fig. 4, the CPMB enables the adaptive sampling of low-frequency categories, facilitating category-guided strong augmentation.

Specifically, the CPMB consists of a set of first-in–first-out (FIFO) queues, one for each category. Each queue stores OLMPs from the labeled dataset. The storage capacity for each category's queue, μ, is a key hyperparameter. Based on our ablation studies (detailed in Section IV-C4), we empirically set the capacity μ to 30 for all categories, as this value provides an optimal balance between maintaining sample diversity and managing computational overhead.

The CPMB is not static, and it is updated at every training iteration. During each iteration, new OLMPs from the current labeled mini-batch are enqueued into their respective categories' FIFO queue. If a queue's size exceeds its capacity of 30, the oldest sample is dequeued. This iterative update mechanism ensures the CPMB remains populated with a diverse and up-to-date set of samples.

To construct a truly adaptive Perturbation strategy, the model's current category preferences must be captured dynamically. Therefore, at the beginning of each training iteration, we re-evaluate these preferences using the PFI. We feed the PFI, I_w, into the current model $F(\cdot)$ to obtain the prediction logits Z^W [as in (5)], which directly reflect the model's instantaneous bias toward certain categories.

From these logits Z^W, we then derive a category probability distribution P^W via the softmax function. Here, P^W represents the normalized prediction probabilities for each category on the PFI. Next, we compute the total predicted probability for each category, S_k, by accumulating these probabilities across all pixels

$$S_k = \sum_{i=1}^{H} \sum_{j=1}^{W} p^W_{i,j,k} \tag{10}$$

where S_k represents the total predicted probability for category k on the PFI I_w, and H and W are the height and width of the input image, respectively.

To balance the contribution of each category during training, we assign higher sampling probabilities to categories with lower predicted proportions. Specifically, we compute the sampling probability for each category k as

$$P_k^{\text{sample}} = \frac{1}{\frac{S_k}{\sum_{j=1}^{K} S_j} + \epsilon} \tag{11}$$

where S_k represents the total predicted probability for category k, K is the total number of categories, and ϵ is a small positive constant to prevent division by zero. To ensure that the total sampling probabilities sum to 1, we normalize P_k^{sample} by dividing each P_k^{sample} by the sum of all sampling probabilities.

Based on the computed category sampling probabilities P_k^{sample}, we perform sampling without replacement in the CPMB to select $K/3$ categories. These selected categories are then randomly shuffled, and according to the shuffled order, we construct the category-filled image I_s and the corresponding category-filled label Y_s from the CPMB for data augmentation.

1) Initialize the category sample image I_s with 0 and the category label Y_s with 255.
2) Guided by the sampling probabilities defined in (11), sequentially select $K/3$ OLMP from the CPMB, where each pair consists of an original image and its corresponding labeled mask.
3) After shuffling, store the category information in I_s and Y_s based on the selected order. Since category samples may overlap, the order of category filling in I_s and Y_s is randomly adjusted to prevent the same category from being repeatedly filled, ensuring effective data augmentation.

The augmented samples x_i^{s1} and labels y_i^{s1} are then constructed as follows:

$$x_i^{s1} = (1 - M) \odot x_i^{u_w} + M \odot I_s \tag{12}$$

$$y_i^{s1} = (1 - M) \odot y_i^{u_w} + M \odot Y_s \tag{13}$$

where x_{iu}^w and y_{iu}^w denote the weakly augmented unlabeled data images and pseudo-labels, respectively, and $\odot$ represents pixel-wise multiplication. M is a binary mask derived from the category-filled label Y_s. Specifically, $M = 1$ indicates positions where the content from the category-filled image I_s and label Y_s is preserved—meaning that, at these locations, the original image x_{iu}^w and pseudo-label y_{iu}^w are replaced with the augmented category information. The final augmented samples

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x_i^{s1} and labels y_i^{s1} are used as a strong augmentation branch in CPSeg, further enhancing the model's learning ability for low-frequency categories and mitigating the training bias induced by the long-tailed distribution.

Algorithm 1 Training Procedure of CPSeg

Require: $D^U = \{x_i^u\}$; $D^L = \{(x_i^l, y_i)\}$
Ensure: Optimized model parameters $\boldsymbol{\theta}$
Initialize CPMB by filling OLMP
while epoch $\leq N_{\text{epoch}}$ **do**
  Compute category weights $\tilde{w}_t^k$ according to Eq. (9)
  for iteration = 1 to $N_{\text{epoch_iter}}$ **do**
    Update CPMB based on current mini-batch
    Compute supervised loss $\mathcal{L}_s$ via Eq. (2)
    $x_i^{u_w}$ = WeakAugment(x_i^u)
    $y_i^{u_w}$ = Model($x_i^{u_w}$)
    $x_i^{s_2}$ = StrongAugment($x_i^{u_w}$)
    $y_i^{s_2}$ = Model($x_i^{s_2}$)
    Generate $x_i^{s_1}$ and $y_i^{s_1}$ via Eqs. (11), (12), and (13)
    Compute losses $\mathcal{L}_{u_1}$ and $\mathcal{L}_{u_2}$ via Eq. (4)
    $\mathcal{L} = \mathcal{L}_s + 0.5 \cdot \mathcal{L}_{u_1} + 0.5 \cdot \mathcal{L}_{u_2}$
    Back propagating the $\mathcal{L}$ and update parameters
  end for
end while
return $\boldsymbol{\theta}$

The overall algorithm process of the CPSeg framework is shown in Algorithm 1.

IV. Experiments

In this section, we design a series of experiments to validate the feasibility and advancement of our method. We first introduce the implementation details, including the datasets, implementation procedures, and evaluation metrics. Subsequently, we conduct comparative experiments using five publicly available semi-supervised semantic segmentation methods and our proposed method on three RS datasets. The evaluation is based on both the performance metrics and visualizations. To further verify the rationale and effectiveness of our semi-supervised framework, we perform ablation experiments on AWACP, APACP, and the dual-perturbation strategy, further demonstrating the effectiveness of these strategies.

A. Dataset

The datasets used in our experiments are from publicly available sources. In this experiment, we used the GID-15 and MSL and Potsdam datasets.

The GID-15 [54] dataset consists of 10 pixel-level annotated images, each with a size of 7200 × 6800 pixels, covering 15 categories, including industrial land, urban residential, rural residential, traffic land, paddy field, irrigated land, dry cropland, garden land, arbor woodland, SL, natural grassland, artificial grassland (AG), river, lake, and pond. This dataset effectively reflects the diversity of the real Earth's surface, with precise annotations that meet the requirements for detailed classification. It is widely used in scene understanding and semantic segmentation research of RS images. Furthermore, the dataset exhibits a long-tailed distribution, with certain categories having significantly fewer samples compared to others.

The MSL-Seg [55] dataset contains 4155 RGB color images collected by the mast camera (MastCam) of the Curiosity Rover (MSL), with the images downsampled to a resolution of 560×500 via bilinear interpolation. The dataset divides the Martian surface terrain into nine categories: Martian soil, sand, bedrock, rocks, unknown regions, gravel, tracks, shadows, and background. These categories contribute to a more comprehensive understanding of Martian landscapes and provide rich annotated data for Martian scene segmentation tasks.

The Potsdam dataset [56] is a widely recognized benchmark for urban semantic segmentation, originating from the ISPRS 2-D Semantic Labeling Contest. It comprises 38 high-resolution true orthophotos of Potsdam, Germany, with each image having a size of 6000 × 6000 pixels and a very high spatial resolution of 5 cm. Each pixel is meticulously annotated into one of six distinct categories: impervious surface, building, low vegetation, tree, car, and clutter. Due to its high level of detail and diverse urban scenes, the Potsdam dataset serves as a standard for evaluating the performance of RS image segmentation algorithms, particularly in complex urban environments.

B. Implementation Details

In all experiments, we built the semi-supervised segmentation model using the PyTorch framework and trained it on an A5000 GPU. The model uses ResNet-101 as the backbone and employs DeepLabV2 as the framework. Key parameters during training include: the use of the SGD optimizer with an initial learning rate of 0.001, momentum of 0.9, and weight decay of 1×10^{-4}. Each batch consists of four unlabeled images and four labeled images, with image cropping to a size of 321×321 pixels. In addition, the queue capacity of the CPMB is set to 30, and the training epochs are set to 100.

C. Ablation Study

In this section, we investigate the effectiveness of the two adaptive strategies proposed in CPSeg, namely, AWACP and APACP. Furthermore, we validate our preference modeling through a dedicated analysis of the PFI and investigate the framework's sensitivity to the annotation rate.

We also assess the impact of key hyperparameters on the experimental results, including the adaptive perturbation weight parameter λ, the adaptive weight update frequency parameter γ, and the memory bank capacity μ.

1) Analysis of Framework Components: To validate the effectiveness of our framework's core components, we conduct a comprehensive ablation study on both the GID-15 and MSL datasets. This analysis first examines the individual contributions of the AWACP and APACP modules. In addition, it investigates the superiority of our proposed PFI mechanism for modeling category preference, which is the cornerstone of both modules. To this end, we construct a strong baseline variant named CPSeg-PL, which retains both adaptive modules but replaces our PFI-based preference modeling with a

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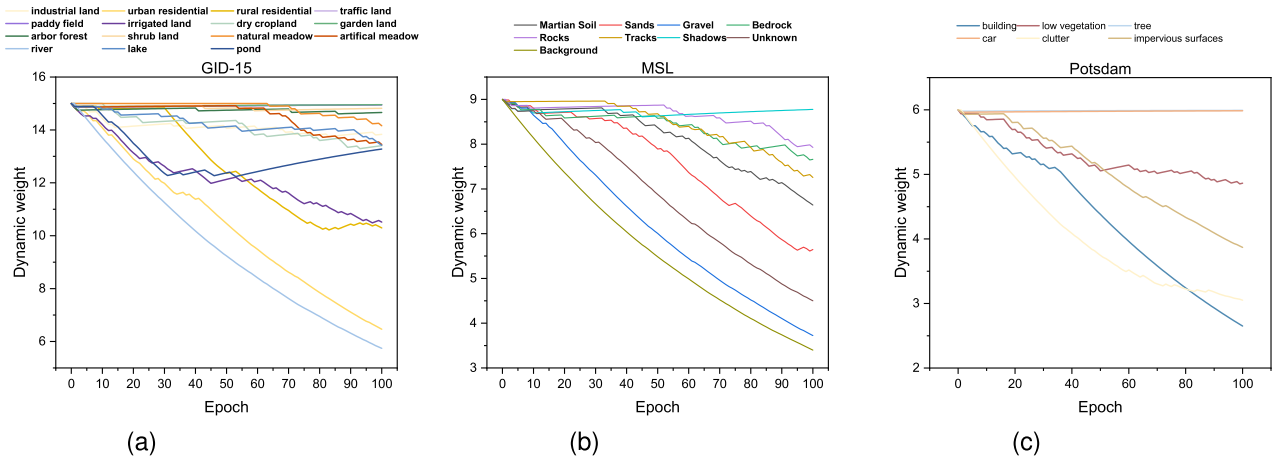


Fig. 5. Weight curves during training on three datasets. (a) Weight curve on the GID-15 dataset with a 1/8 labeled data ratio. (b) Weight curve on the MSL dataset with a 1/8 labeled data ratio. (c) Weight curve on the Potsdam dataset with a 1/8 labeled data ratio.

TABLE I

ABLATION STUDY RESULTS EVALUATING THE EFFECTIVENESS OF EACH STRATEGY ON THE GID-15 DATASET WITH A 1/8 LABELED DATA RATIO. BOTH MIOU AND PER CATEGORY IOU ARE REPORTED

Method	per category IoU															mIoU
	InL	UR	RR	TL	PF	IrL	DC	GL	AW	SL	NG	AG	Riv.	Lak.	Pon.	
CPSeg	71.63	78.48	70.98	67.34	80.67	90.43	80.29	22.70	97.48	56.22	92.15	68.64	87.12	84.06	69.38	**74.50**
CPSeg-PL	71.60	78.59	70.17	67.18	73.19	90.21	78.23	22.90	97.41	54.66	91.02	68.82	87.38	84.31	63.93	73.31
w/o AWACP	70.78	78.39	69.84	66.58	80.90	89.73	77.57	20.22	97.31	48.87	88.89	65.83	86.60	84.44	70.45	73.09
w/o APACP	71.18	77.72	69.93	66.96	80.71	90.19	78.03	22.29	97.52	55.47	91.94	68.87	86.93	84.30	70.18	74.15
w/o AWACP&APACP	71.08	78.40	69.93	66.28	79.86	89.53	77.48	20.53	97.29	42.96	89.16	66.51	87.06	83.23	70.51	72.65

TABLE II

ABLATION STUDY RESULTS EVALUATING THE EFFECTIVENESS OF EACH STRATEGY ON THE MSL DATASET WITH A 1/8 LABELED DATA RATIO. BOTH MIOU AND PER CATEGORY IOU ARE REPORTED

Method	per category IoU									mIoU
	Martian soil	Sand	Bedrock	Rocks	Unknown	Gravel	Tracks	Shadows	Background	
CPSeg	28.13	65.24	74.80	41.24	58.80	69.09	65.99	42.33	87.76	**59.26**
CPSeg-PL	27.45	64.31	74.50	38.97	56.97	68.77	57.21	41.21	87.41	57.42
w/o AWACP	26.73	64.56	74.65	39.76	57.76	69.28	65.21	41.75	87.90	58.62
w/o APACP	27.62	65.33	75.14	39.80	58.23	69.56	64.86	42.06	87.39	58.89
w/o AWACP&APACP	25.34	65.11	73.86	39.61	57.98	68.07	64.03	41.97	87.64	58.18

simpler estimation derived from the statistical distribution of pseudo-labels. By comparing CPSeg against CPSeg-PL, we can thoroughly assess our design choices.

As shown in Tables I and II, under the 1/8 labeled data setting, removing either AWACP or APACP results in noticeableperformance degradation. The removal of AWACP, for instance, causes the mean intersection over union (mIoU) to drop from 74.50% to 73.09% on GID-15. This highlights the importance of its adaptive weighting mechanism, which, as visualized in Fig. 5, counteracts learning bias by progressively lowering the weights for well-performing head categories while sustaining high weights for challenging tail classes throughout the training process. Similarly, removing APACP leads to a performance drop, and removing both modules results in the most significant decline to 72.65% on GID-15 and 58.18% on MSL. These results quantitatively confirm the critical and complementary roles of both modules.

Furthermore, the comparison with CPSeg-PL reveals a crucial insight into why our method is effective. Our PFI-based method (CPSeg) consistently and significantly outperforms the pseudo-label-based alternative (CPSeg-PL). On the GID-15 dataset with a 1/8 labeled data setting (Table I), CPSeg achieves an mIoU of 74.50%, surpassing CPSeg-PL (73.31%) by a large margin of 1.19%. This demonstrates that in low-data regimes where pseudo-labels are inherently noisy, our PFI-based preference estimation provides a much more stable and reliable signal for guiding the model. Even when the labeled data is increased to 1/4 (Table III), and CPSeg (76.21%) still maintains a clear advantage over CPSeg-PL (75.10%).

These results clearly demonstrate that the effectiveness of our framework stems not only from the adaptive strategies themselves but, more fundamentally, from the robust PFI-based preference modeling that drives them. By decoupling preference estimation from the potentially flawed supervisory

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TABLE III

ABLATION STUDY RESULTS EVALUATING THE EFFECTIVENESS OF EACH STRATEGY ON THE GID-15 DATASET WITH A 1/4 LABELED DATA RATIO. BOTH MIOU AND PER CATEGORY IOU ARE REPORTED

Method	per category IoU															mIoU
	InL	UR	RR	TL	PF	IrL	DC	GL	AW	SL	NG	AG	Riv.	Lak.	Pon.	
CPSeg	72.80	79.76	72.14	68.44	81.99	91.90	81.60	30.43	99.07	57.14	93.65	69.76	88.54	85.43	70.51	**76.21**
CPSeg-PL	73.00	79.15	71.11	67.89	80.14	90.79	79.57	28.12	98.68	54.72	93.24	69.06	87.54	84.69	68.73	75.10
w/o AWACP	72.21	79.09	70.66	67.69	82.37	90.13	78.25	28.79	98.15	49.23	91.81	67.47	88.66	84.64	73.38	74.84
w/o APACP	72.59	79.56	70.87	68.03	81.78	90.88	78.75	29.93	98.64	55.98	92.85	68.23	88.86	84.71	73.06	75.65
w/o AWACP&APACP	71.76	79.45	70.62	67.48	81.57	90.63	78.16	28.64	98.57	43.73	92.16	67.83	89.03	84.15	73.47	74.48

TABLE IV

COMPARISON OF DIFFERENT PFIS UNDER VARYING LABEL RATIOS ON GID-15

Type of PFI	1/8 Labeled (mIoU)	1/4 Labeled (mIoU)
Black image (0.0 brightness)	73.35	75.18
Gray image (0.2 brightness)	73.80	75.52
Gray image (0.5 brightness)	73.20	75.27
Random noise	72.21	74.38
Gaussian noise	73.28	75.25
Red image	**74.60**	76.09
Yellow image	74.38	76.12
White image	74.50	**76.21**

signal of pseudo-labels, CPSeg achieves a more reliable mechanism for addressing the problem of long-tailed distribution.

However, we note that gains on certain "hard" tail classes, like garden land (Table I), are modest. This is attributable to a dual challenge of extreme data scarcity and feature ambiguity, which can limit the effectiveness of APACP's augmentations for such visually indistinct categories.

In summary, these results demonstrate that our framework's effectiveness stems not only from the adaptive strategies but, more fundamentally, from the robust PFI-based preference modeling that underpins them. By decoupling preference estimation from noisy pseudo-labels, the integration of AWACP and APACP creates a balanced and effective framework for tackling long-tailed distributions in RS.

2) Analysis of PFI: A core assumption of our method is that a PFI can effectively probe the model's intrinsic category preferences without introducing visual information like texture or shape. However, this assumption warrants careful validation, as a single type of PFI (e.g., an all-white image) might inadvertently correspond to a specific color pattern. Therefore, we conducted a comprehensive ablation study to validate our design choice and establish a more general principle for selecting an optimal PFI. We tested a variety of inputs, including solid colors (black, gray, red, yellow, and white), random noise, and Gaussian noise.

The results, presented in Table IV, lead to two key conclusions. First, solid-color inputs consistently outperform noise-based alternatives (e.g., random noise yields the lowest mIoU). This suggests that while noise patterns lack high-level semantic objects, their low-level statistical variations act as a distraction, interfering with the model's ability to reveal its pure category biases.

Second, the choice among solid colors is critical. We observe a clear performance gap between black and the other solid colors. This phenomenon is attributable to a more nuanced definition of an effective PFI. An ideal PFI must satisfy two conditions simultaneously: it must have high uniformity to avoid distracting the model with complex textures and shapes, and it must have low semantic correlation to avoid strong, learned associations with any major categories in the dataset.

The GID-15 dataset contains prevalent features like shadows and deep water bodies that are typically very dark, often approaching pure black. Consequently, a uniform black image possesses a strong semantic correlation that biases the model, making it a poor choice for probing a truly "unbiased" state. In contrast, while the "industrial land" category does contain light-colored pixels, these are characteristically textured and high-entropy, rather than a uniform white expanse. Thus, a pure white PFI's high uniformity and low semantic correlation with a specific, well-defined class in GID-15 allow it to serve as a more effective probe than a black image.

This analysis validates that the success of our framework is not tied to a specific color but to the robust principle of using a semantically-unbiased and uniform input. Our experiments show that red, yellow, and white images are all effective choices for the GID-15 dataset, with the white image being our default implementation due to its combination of simplicity, high uniformity, and minimal semantic correlation.

3) Sensitivity to Annotation Rate: To further deepen our analysis, we investigate the sensitivity of our adaptive strategies to the annotation rate by comparing their performance contributions under 1/8 and 1/4 labeled settings, as detailed in Tables I and II.

Our results reveal that the APACP strategy is notably more sensitive to the annotation rate than AWACP. As the labeled ratio increases from 1/8 to 1/4, the mIoU gain from APACP rises from 0.35% to 0.56%. In contrast, the contribution from AWACP remains highly stable, changing minimally from 1.41% to 1.37%.

This disparity stems from their distinct operational mechanisms. The heightened sensitivity of APACP is due to its direct dependence on labeled data. A higher annotation rate enriches its CPMB with a more diverse and high-quality pool of ground-truth samples. This allows APACP to generate more effective category-guided augmentations, leading to a larger performance gain as supervision increases. Conversely, AWACP's dependence is indirect, as it probes the model's learned category preferences rather than directly consuming labeled samples. While more labeled data yields a better-trained model with more stable preferences, this benefit is mediated through the model's state. Consequently,

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TABLE V

QUANTITATIVE RESULTS OF CPSEG AND STATE-OF-THE-ART SEMI-SUPERVISED SEMANTIC SEGMENTATION METHODS ON LABELED GID-15 DATASET. BOTH MIOU AND PER CATEGORY IOU ARE REPORTED

Ratio	Method	per category IoU															mIoU
		InL	UR	RR	TL	PF	IrL	DC	GL	AW	SL	NG	AG	Riv.	Lak.	Pon.	
1/8	LSST	69.82	78.24	68.64	66.71	75.27	88.77	69.75	38.61	97.83	11.74	91.33	63.89	85.49	84.51	71.82	70.83
	WSCL	73.78	80.34	73.16	71.40	82.27	90.38	73.15	20.22	97.27	45.02	92.23	62.87	85.25	85.59	71.91	73.66
	Fixmatch	69.77	78.23	68.56	65.64	82.12	89.69	75.17	18.62	97.34	1.08	91.21	55.08	87.88	82.61	72.31	69.02
	Unimatch	68.84	77.87	69.25	64.02	81.49	89.61	74.33	37.92	97.55	25.98	90.91	57.29	91.93	85.31	75.11	72.49
	Allspark	69.29	76.75	68.00	64.17	80.36	88.90	66.58	24.20	97.73	34.12	89.86	65.00	82.20	82.80	68.53	70.57
	DWL	69.12	77.29	69.64	65.80	81.20	89.82	72.97	33.59	97.89	44.28	91.28	60.21	85.56	80.31	67.57	72.44
	CPSeg	71.63	78.48	70.98	67.34	80.67	90.43	80.29	22.70	97.48	56.22	92.15	68.64	87.12	84.06	69.38	**74.50**
1/4	LSST	69.72	78.85	70.35	70.42	80.80	89.92	77.50	42.12	97.29	26.07	92.57	60.31	89.90	81.65	73.52	73.40
	WSCL	72.66	79.63	70.89	74.28	81.11	90.75	79.96	39.65	98.33	19.88	92.81	63.50	90.24	82.69	74.70	74.07
	Fixmatch	70.90	78.78	70.02	68.05	81.01	90.56	81.13	40.00	98.74	42.61	93.33	57.65	90.79	84.43	72.94	74.73
	Unimatch	72.37	78.32	69.69	67.24	82.63	90.58	81.30	39.08	98.52	54.62	92.69	62.42	92.07	78.84	68.38	75.25
	Allspark	68.85	77.07	68.53	64.88	83.09	90.18	79.37	32.73	96.91	43.92	92.17	61.01	85.10	83.39	70.74	73.20
	DWL	70.40	78.77	69.87	65.74	81.72	90.64	80.18	47.16	98.52	64.72	90.97	62.41	88.68	79.41	72.15	76.09
	CPSeg	72.80	79.76	72.14	68.44	81.99	91.90	81.60	30.43	99.07	57.14	93.65	69.76	88.54	85.43	70.51	**76.21**

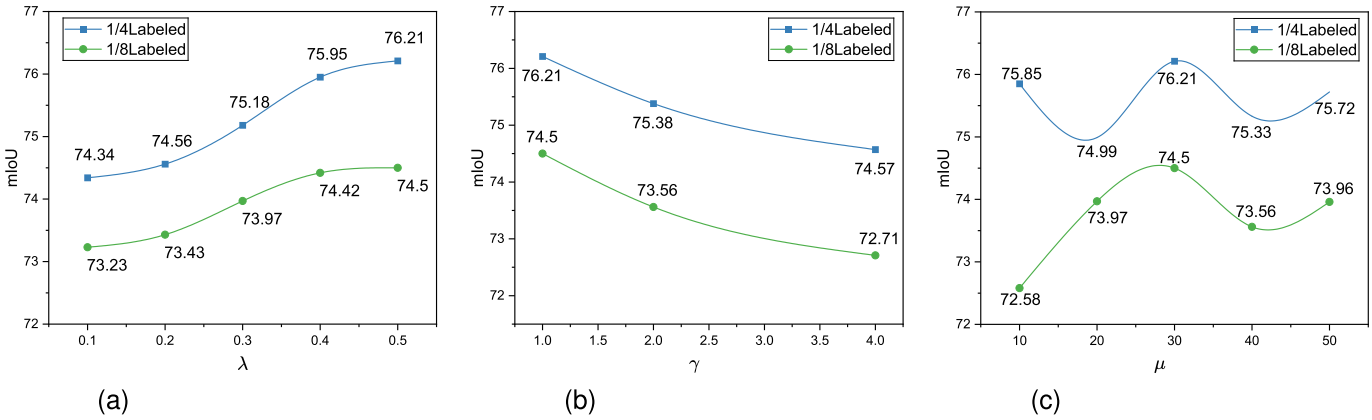


Fig. 6. Parameter sensitivity analysis on the GID-15 dataset. (a) Influence of parameter α on mIoU. (b) Influence of parameter γ on mIoU. (c) Influence of parameter μ on mIoU.

TABLE VI

QUANTITATIVE RESULTS OF CPSEG AND STATE-OF-THE-ART SEMI-SUPERVISED SEMANTIC SEGMENTATION METHODS ON MSL DATASET. BOTH MIOU AND PER CATEGORY IOU ARE REPORTED

Ratio	Method	per category IoU									mIoU
		Martian soil	Sand	Bedrock	Rocks	Unknown	Gravel	Tracks	Shadows	Background	
1/8	LSST	22.83	57.82	70.47	30.73	51.58	65.45	45.85	33.40	85.33	51.50
	WSCL	24.29	64.39	74.80	39.69	58.24	68.45	69.01	40.47	86.06	58.38
	Fixmatch	26.46	65.57	75.07	38.72	52.65	68.92	66.39	41.26	87.44	58.05
	Unimatch	29.33	66.13	75.55	42.33	56.77	69.61	58.71	42.39	87.66	58.72
	Allspark	22.18	67.34	71.23	40.63	61.31	69.58	51.30	41.13	86.01	56.75
	DWL	31.35	65.47	72.62	39.02	49.79	69.76	60.54	39.52	88.49	57.40
	CPSeg	28.13	65.24	74.80	41.24	58.80	69.09	65.99	42.33	87.76	**59.26**
1/4	LSST	32.35	61.62	74.54	37.16	51.69	70.09	50.30	43.80	87.80	56.60
	WSCL	30.77	65.53	75.48	40.90	63.99	70.24	66.84	42.84	89.02	60.62
	Fixmatch	32.91	61.61	76.03	44.83	59.78	70.20	71.14	44.96	87.66	61.01
	Unimatch	33.48	65.23	76.53	43.70	62.75	70.42	67.28	45.99	89.57	61.66
	Allspark	35.48	68.27	76.27	44.86	61.14	71.62	54.07	47.65	88.97	60.93
	DWL	30.50	65.00	74.00	37.40	63.40	69.50	54.80	43.90	89.10	58.62
	CPSeg	34.35	65.96	76.74	42.32	62.45	71.59	70.44	46.35	88.77	**62.11**

its performance is coupled with the quality of the learned representation, not the raw quantity of labels, resulting in less pronounced sensitivity to the annotation rate. This analysis highlights the complementary nature of our strategies in leveraging different signals within the semi-supervised framework.

4) Sensitivity Analysis of Parameters λ, γ, and μ: To further investigate the robustness and sensitivity of our proposed adaptive strategies, we conduct a parameter sensitivity analysis on the GID-15 dataset by varying the values of λ, γ, and μ, which control the weight of the adaptive perturbation branch,

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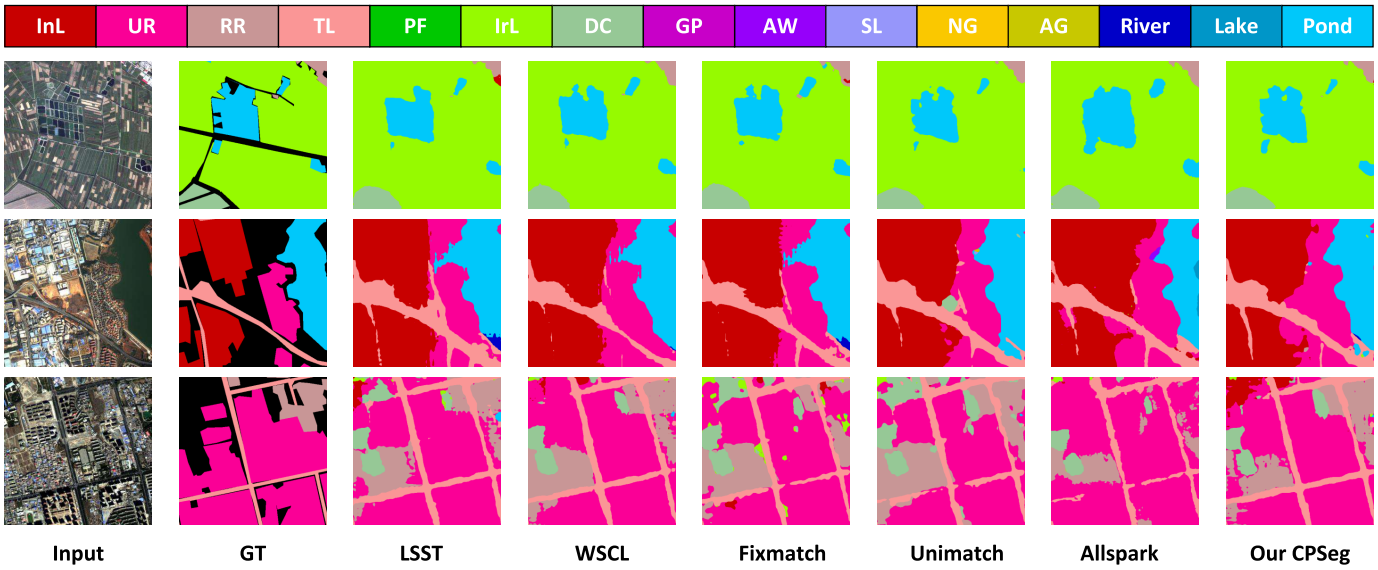


Fig. 7. Visual results of different methods on the GID-15 dataset with a 1/4 labeled data ratio.

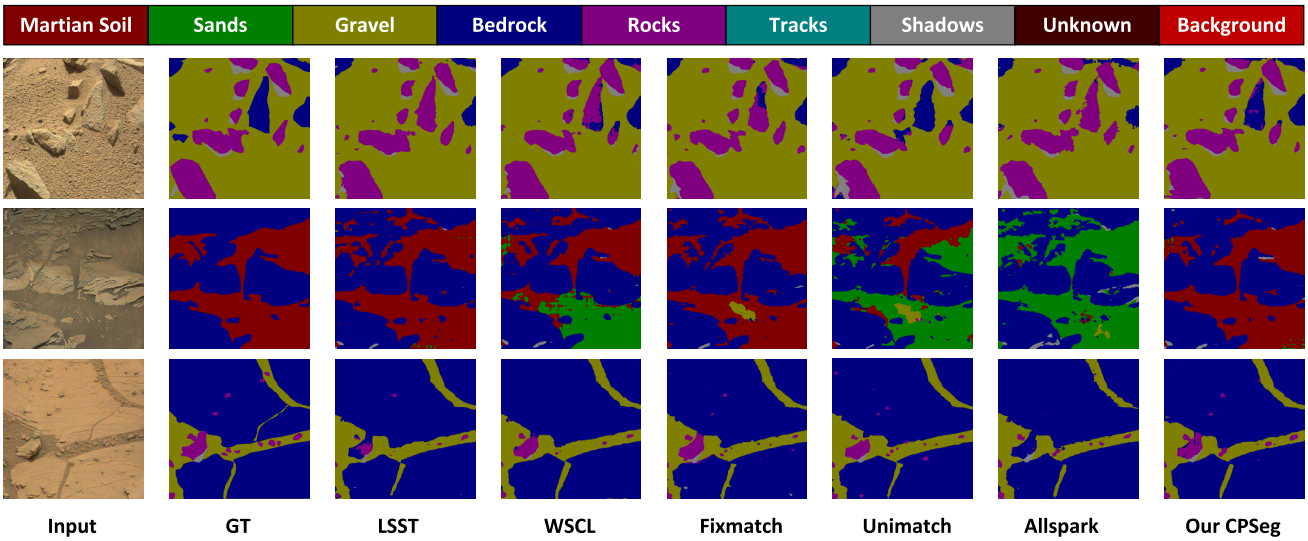


Fig. 8. Visual results of different methods on the MSL dataset with a 1/4 labeled data ratio.

the frequency of adaptive weight updates, and the capacity of the CPMB, respectively.

In our dual-stream perturbation design, the strong augmentation is applied through two parallel branches—one using standard strategies and the other employing our adaptive perturbation method. The parameter λ controls the contribution of the adaptive branch in the semi-supervised loss, with the complementary weight $1 - \lambda$ assigned to the standard augmentation branch. As shown in Fig. 6(a), increasing λ from 0.1 to 0.5 leads to consistent improvements in mIoU under both 1/4 and 1/8 labeled data settings. For example, the mIoU on the 1/4 setting improves from 74.34% to 76.21%, and from 73.23% to 74.50% under the 1/8 setting. These results indicate that our adaptive augmentation stream effectively promotes the learning of challenging or low-confidence samples. A higher λ gives the model more freedom to explore informative decision boundaries via stronger and category-aware perturbations, which is especially beneficial under limited supervision.

The parameter γ controls how frequently the category weights in the AWACP module are updated during training. We evaluate the impact of updating these weights every 1, 2, or 4 epochs. As illustrated in Fig. 6(b), a smaller update interval (i.e., more frequent updates) generally leads to better performance. Specifically, updating every 1 epoch achieves the best performance (76.21% and 74.50% on the 1/4 and 1/8 settings, respectively), while increasing the interval to 4 epochs leads to performance drops (down to 74.57% and 72.71%). This trend suggests that dynamically capturing the evolving category difficulty and preference distribution is crucial for effectively guiding the optimization process. Less frequent updates may cause the model to rely on outdated or suboptimal weight distributions, reducing the effectiveness of the AWACP module in addressing the long-tailed distribution and difficult samples.

As illustrated in Fig. 6(c), the capacity μ of the CPMB queue governs a critical tradeoff between capturing suf-

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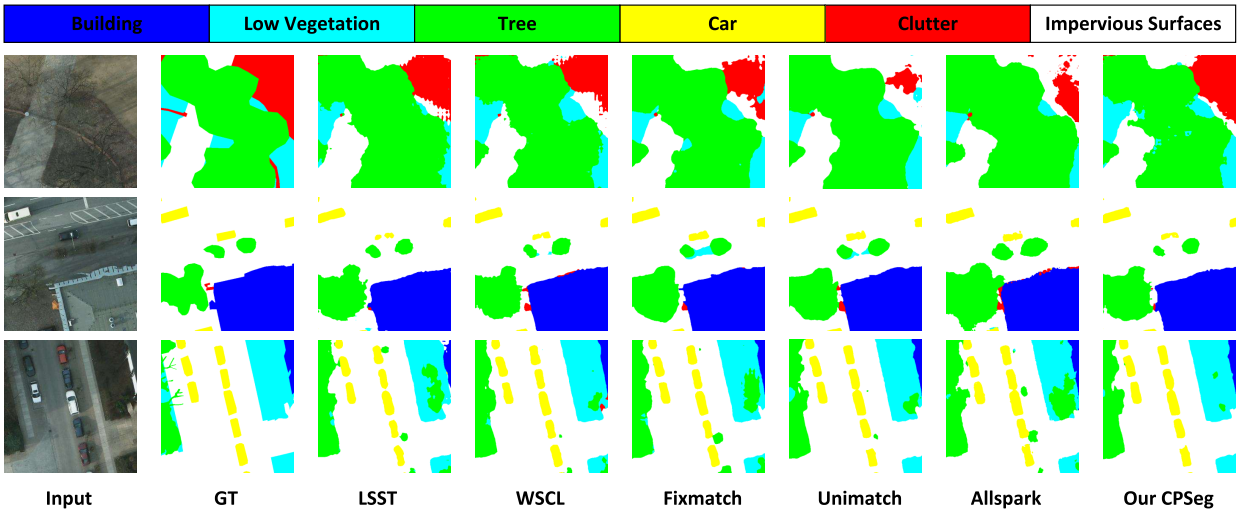


Fig. 9. Visual results of different methods on the Potsdam dataset with a 1/4 labeled data ratio.

ficient intraclass diversity and mitigating "sliding window solidification"—our term for the model overfitting to a slowly-updated set of exemplars due to their high sampling rate by APACP. Our analysis reveals that the optimal μ balances the window's representativeness against its update dynamism.

A small capacity like $\mu = 10$ creates a highly dynamic but potentially biased window. Under the 1/4 setting, this high dynamism (i.e., fast sliding speed) effectively prevents overfitting, leading to strong performance (75.85%). However, in the data-scarce 1/8 setting, the same small window fails to capture enough diversity from the already limited pool of exemplars, and its update dynamism is also comparatively slower, thus constraining performance to just 72.58%. Conversely, an excessively large μ (e.g., $\geq$40) slows the window's update speed across the board, increasing the residence time of each sample. This amplifies solidification risk and harms generalization in both settings, causing performance to decline.

The nonmonotonic trend, especially the performance drop to 74.99% mIoU at $\mu = 20$ in the 1/4 setting, underscores this delicate balance. Here, the window becomes slower, increasing solidification risk, yet it is still too small to be a sufficiently representative sample of the class's broader diversity. Ultimately, $\mu = 30$ emerges as the most robust choice, achieving peak performance in both scenarios. It is large enough to significantly enhance the diversity captured in the data-scarce 1/8 setting (boosting mIoU to 74.50%), while striking an optimal balance between representativeness and dynamism in the data-richer 1/4 setting (reaching 76.21%).

Overall, as demonstrated in Fig. 6, the parameter configuration $\lambda = 0.5$, $\gamma = 1$, and $\mu = 30$ offers the most robust and effective performance across both labeled data settings. This indicates that a full contribution from the adaptive perturbation branch, frequent updates of adaptive weights, and a well-balanced memory capacity that robustly balances diversity capture and update dynamism collectively create the most effective framework for semi-supervised segmentation under limited supervision.

D. Comparison With State-of-the-Art Methods

In this section, we compare our method with several popular semi-supervised semantic segmentation approaches, including those widely used in natural scene understanding, as well as methods that are particularly popular in the field of RS. To ensure fair comparisons, all experiments are conducted using a single NVIDIA A5000 GPU, with training consistently run for 100 epochs across all methods.

1) Comparative Analysis on the GID-15 Dataset: On the GID-15 dataset, CPSeg demonstrates significant superiority under both 1/8 and 1/4 labeled ratios, as shown in Table V. With only 1/8 of labels, CPSeg achieves a state-of-the-art mIoU of 74.50%, surpassing the previous best method, WSCL (73.66%), by 0.84%. This performance gain is particularly evident in challenging long-tailed categories. For instance, in the SLs category, CPSeg boosts the IoU from 45.02% (WSCL) to 56.22%, showcasing its enhanced ability to extract supervisory signals from sparse annotations. When the labeled ratio increases to 1/4, our method's mIoU further rises to 76.21%, with the IoU for AGs outperforming WSCL by 6.26%. The qualitative results in Fig. 7 show that while most methods capture the main structures, CPSeg produces results with finer detail refinement, validating its effectiveness in real-world RS scenarios with long-tailed distributions.

2) Comparative Analysis on the Msl Dataset: To evaluate CPSeg's efficacy in planetary terrain segmentation, we conducted experiments on the MSL dataset. The results in Table VI indicate that CPSeg consistently outperforms all baseline methods. Under the 1/8 label setting, it achieves a mIoU of 59.26%, and this increases to 62.11% with 1/4 of labels. While the overall mIoU improvement is moderate, CPSeg shows notable advantages in challenging categories characterized by irregular boundaries and small object sizes. This suggests a strong capability in capturing fine-grained features under limited supervision. The visualizations in Fig. 8 further highlight this strength. In scenes with high category sparsity and complex backgrounds where other methods falter, CPSeg more effectively preserves the information of small

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TABLE VII
QUANTITATIVE RESULTS OF CPSEG AND STATE-OF-THE-ART SEMI-SUPERVISED SEMANTIC SEGMENTATION METHODS ON POTSDAM DATASET. BOTH MIOU AND PER CATEGORY IOU ARE REPORTED

Ratio	Method	per category IoU						mIoU
		Building	Low Veg.	Tree	Car	Clutter	Imp. Surf.	
1/8	LSST	78.28	64.72	66.13	58.58	40.53	71.31	63.26
	WSCL	83.24	68.93	70.62	63.32	45.75	75.88	67.96
	Fixmatch	83.43	68.63	69.21	64.50	45.15	76.17	67.85
	Unimatch	83.34	69.17	69.04	64.14	43.20	75.31	67.37
	Allspark	82.69	67.48	69.36	64.90	38.38	75.12	66.32
	DWL	82.12	68.00	69.12	65.64	43.13	75.14	67.19
	CPSeg	83.41	68.90	70.56	65.45	45.42	76.05	**68.30**
1/4	LSST	81.44	67.89	68.89	59.59	47.25	74.33	66.57
	WSCL	83.88	70.52	71.06	61.71	50.67	77.12	69.16
	Fixmatch	84.86	69.76	70.02	64.59	49.34	77.29	69.31
	Unimatch	84.47	69.89	69.58	64.68	48.95	77.16	69.12
	Allspark	82.70	68.49	69.82	65.53	46.30	75.95	68.13
	DWL	82.47	68.77	69.85	65.49	48.33	75.97	68.48
	CPSeg	84.59	70.51	71.51	65.62	49.72	77.20	**69.86**

and discrete categories, demonstrating its practical potential for planetary exploration tasks.

3) Comparative Analysis on the Potsdam Dataset: On the Potsdam urban semantic segmentation dataset, CPSeg achieves competitive performance, as detailed in Table VII. With a 1/8 labeled ratio, our method reaches a mIoU of 68.30%, which improves to 69.86% at a 1/4 ratio. The performance gains are particularly noticeable in often-confused categories, such as car, clutter, and tree. This observation is supported by the sustained high weights of the tree and car categories during training, as visualized in Fig. 5(c), indicating that our model effectively focuses on these intricate urban features. The visual results in Fig. 9 reveal that while most methods perform adequately on large objects with clear boundaries, their performance on smaller objects is less distinct. In contrast, CPSeg exhibits superior detail preservation, especially for cars and trees, highlighting its effectiveness in handling the complex characteristics of urban environments under SSL.

V. CONCLUSION

In this work, we propose CPSeg, a novel framework that explicitly models the model's category preference to address the challenges posed by long-tailed distribution and limited labeled data in RS image segmentation. By feeding PFI into the model, CPSeg captures its preference toward different categories without relying on pseudo-labels. This learned category preference information plays a central role in two proposed strategies. The AWACP dynamically emphasizes low-frequency categories, promoting interclass balance. APACP leverages category preference information and updates the OLMP in the CPMB to guide category-aware sampling, encouraging augmented samples to favor tail categories, thereby mitigating the adverse effects of the long-tailed distribution. Experiments on the GID-15, MSL, and Potsdam datasets demonstrate that CPSeg outperforms existing methods. It effectively enhances the model's ability to recognize minority categories, improves the use of unlabeled data, and promotes overall generalization. Future work will focus on a more precise modeling and utilization of category preference, particularly to address the challenge of "hard" tail categories where scarcity and ambiguity coexist. Promising directions include exploring advanced generative techniques to enrich training data and developing more discriminative feature learning strategies to better separate confusing classes. Furthermore, the underlying principles of CPSeg show great promise for extension to other complex RS tasks such as object detection and change detection.

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REFERENCES

[1] C. Persello et al., "Deep learning and Earth observation to support the sustainable development goals: Current approaches, open challenges, and future opportunities," *IEEE Geosci. Remote Sens. Mag., Replaces Newslett.*, vol. 10, no. 2, pp. 172–200, Jun. 2022.

[2] J. Yue et al., "Optical remote sensing image understanding with weak supervision: Concepts, methods, and perspectives," *IEEE Geosci. Remote Sens. Mag. Replaces Newslett.*, vol. 10, no. 2, pp. 250–269, Jun. 2022.

[3] M. Maboudi, J. Amini, S. Malihi, and M. Hahn, "Integrating fuzzy object based image analysis and ant colony optimization for road extraction from remotely sensed images," *ISPRS J. Photogramm. Remote Sens.*, vol. 138, pp. 151–163, Apr. 2018.

[4] X. Liu et al., "Classifying urban land use by integrating remote sensing and social media data," *Int. J. Geographical Inf. Sci.*, vol. 31, no. 8, pp. 1675–1696, Aug. 2017.

[5] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117.

[6] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, May 2015, pp. 3431–3440.

[7] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[8] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.

[9] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[10] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking Atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[11] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[12] M. Cordts et al., "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3213–3223.

[13] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1195–1204.

[14] L. Yang, W. Zhuo, L. Qi, Y. Shi, and Y. Gao, "ST++: Make self-training work better for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4268–4277.

[15] G. French, S. Laine, T. Aila, M. Mackiewicz, and G. Finlayson, "Semi-supervised semantic segmentation needs strong, high-dimensional perturbations," 2019, *arXiv:1906.01916*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:22:37 UTC from IEEE Xplore. Restrictions apply.

[16] Y. Chen, X. Ouyang, K. Zhu, and G. Agam, "ComplexMix: Semi-supervised semantic segmentation via mask-based data augmentation," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2021, pp. 2264–2268.
[17] S. Yun, D. Han, S. Chun, S. J. Oh, Y. Yoo, and J. Choe, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6023–6032.
[18] V. Olsson, W. Tranheden, J. Pinto, and L. Svensson, "ClassMix: Segmentation-based data augmentation for semi-supervised learning," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, Jan. 2021, pp. 1369–1378.
[19] Y. Ouali, C. Hudelot, and M. Tami, "Semi-supervised semantic segmentation with cross-consistency training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12674–12684.
[20] G. French, S. Laine, T. Aila, M. Mackiewicz, and G. Finlayson, "Semi-supervised semantic segmentation needs strong, varied perturbations," 2019, *arXiv:1906.01916*.
[21] X. Chen, Y. Yuan, G. Zeng, and J. Wang, "Semi-supervised semantic segmentation with cross pseudo supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2613–2622.
[22] Q. Xie, M.-T. Luong, E. Hovy, and Q. V. Le, "Self-training with noisy student improves ImageNet classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Sep. 2020, pp. 10687–10698.
[23] J. Yuan, Y. Liu, C. Shen, Z. Wang, and H. Li, "A simple baseline for semi-supervised semantic segmentation with strong data augmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8229–8238.
[24] Z. Feng et al., "DMT: Dynamic mutual training for semi-supervised learning," *Pattern Recognit.*, vol. 130, Oct. 2022, Art. no. 108777.
[25] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2020, pp. 596–608.
[26] Y. Liu, Y. Tian, Y. Chen, F. Liu, V. Belagiannis, and G. Carneiro, "Perturbed and strict mean teachers for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4258–4267.
[27] L. Yang, L. Qi, L. Feng, W. Zhang, and Y. Shi, "Revisiting weak-to-strong consistency in semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7236–7246.
[28] B. Sun, Y. Yang, L. Zhang, M.-M. Cheng, and Q. Hou, "CorrMatch: Label propagation via correlation matching for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3097–3107.
[29] Z. Yang et al., "Label propagation and contrastive regularization for semisupervised semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5609818, doi: 10.1109/TGRS.2023.3277203.
[30] L. Zhang, Z. Tan, Y. Zheng, G. Zhang, W. Zhang, and Z. Li, "A bias correction semi-supervised semantic segmentation framework for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5604414, doi: 10.1109/TGRS.2024.3521420.
[31] X. Lu et al., "Weak-to-strong consistency learning for semisupervised image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510715, doi: 10.1109/TGRS.2023.3272552.
[32] L. Lv and L. Zhang, "Advancing data-efficient exploitation for semi-supervised remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5619213, doi: 10.1109/TGRS.2024.3388199.
[33] R. Liu, T. Luo, S. Huang, Y. Wu, Z. Jiang, and H. Zhang, "CrossMatch: Cross-view matching for semi-supervised remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5650515, doi: 10.1109/TGRS.2024.3507050.
[34] Z. Chen, R. Zhang, G. Zhang, Z. Ma, and T. Lei, "Digging into pseudo label: A low-budget approach for semi-supervised semantic segmentation," *IEEE Access*, vol. 8, pp. 41830–41837, 2020.
[35] H. Li and H. Zheng, "A residual correction approach for semi-supervised semantic segmentation," in *Proc. 4th Chin. Conf. Pattern Recognit. Comput. Vis.*, Beijing, China. Cham, Switzerland: Springer, Oct. 2021, pp. 90–102.
[36] S. Mittal, M. Tatarchenko, and T. Brox, "Semi-supervised semantic segmentation with high- and low-level consistency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1369–1379, Apr. 2021.
[37] Z. Ke, D. Qiu, K. Li, Q. Yan, and R. W. Lau, "Guided collaborative training for pixel-wise semi-supervised learning," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 429–445.
[38] H. Wang, Q. Zhang, Y. Li, and X. Li, "AllSpark: Reborn labeled features from unlabeled in transformer for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3627–3636.
[39] Y. Wang et al., "Semi-supervised semantic segmentation using unreliable pseudo-labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4248–4257.
[40] X. Lu et al., "Simple and efficient: A semisupervised learning framework for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5543516, doi: 10.1109/TGRS.2022.3220755.
[41] J. Jin, W. Lu, H. Yu, X. Rong, X. Sun, and Y. Wu, "Dynamic and adaptive self-training for semi-supervised remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5639814, doi: 10.1109/TGRS.2024.3407142.
[42] Y. Xin, Z. Fan, X. Qi, Y. Zhang, and X. Li, "Confidence-weighted dual-teacher networks with biased contrastive learning for semi-supervised semantic segmentation in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614416, doi: 10.1109/TGRS.2024.3376352.
[43] S. Wang, X. Sun, C. Chen, D. Hong, and J. Han, "Semi-supervised semantic segmentation for remote sensing images via multi-scale uncertainty consistency and cross-teacher–student attention," 2025, *arXiv:2501.10736*.
[44] Y. Luo, B. Sun, S. Li, and Y. Hu, "Hierarchical augmentation and region-aware contrastive learning for semi-supervised semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4401311, doi: 10.1109/TGRS.2024.3521586.
[45] X. Gan, W. Li, Y. Zhang, W. Long, Y. Lu, and Z. Chen, "Prior information-guided semi-supervised semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4411016, doi: 10.1109/TGRS.2025.3571066.
[46] J. Kim, Y. Hur, S. Park, E. Yang, S. J. Hwang, and J. Shin, "Distribution aligning refinery of pseudo-label for imbalanced semi-supervised learning," in *Proc. NIPS*, vol. 33, 2020, pp. 14567–14579.
[47] C. Wei, K. Sohn, C. Mellina, A. Yuille, and F. Yang, "CReST: A class-rebalancing self-training framework for imbalanced semi-supervised learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10857–10866.
[48] H. Peng, W. Pian, M. Sun, and P. Li, "Dynamic re-weighting for long-tailed semi-supervised learning," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 6453–6463.
[49] T. Wei, Q.-Y. Liu, J.-X. Shi, W.-W. Tu, and L.-Z. Guo, "Transfer and share: Semi-supervised learning from long-tailed data," *Mach. Learn.*, vol. 113, no. 4, pp. 1725–1742, Apr. 2024.
[50] H. Wang and X. Li, "DHC: Dual-debiased heterogeneous co-training framework for class-imbalanced semi-supervised medical image segmentation," in *Proc. Int. Conf. Med. image Comput. Comput.-Assist. Intervent*. Cham, Switzerland: Springer, 2023, pp. 582–591.
[51] W. Huang, Y. Shi, Z. Xiong, and X. X. Zhu, "AdaptMatch: Adaptive matching for semisupervised binary segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5625416, doi: 10.1109/TGRS.2023.3332490.
[52] W. Huang, Y. Shi, Z. Xiong, and X. X. Zhu, "Decouple and weight semi-supervised semantic segmentation of remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 212, pp. 13–26, Jun. 2024.
[53] H. Lee and H. Kim, "CDMAD: Class-distribution-mismatch-aware debiasing for class-imbalanced semi-supervised learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 23891–23900.
[54] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.
[55] J. Li, S. Zi, R. Song, Y. Li, Y. Hu, and Q. Du, "A stepwise domain adaptive segmentation network with covariate shift alleviation for remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5618515, doi: 10.1109/TGRS.2022.3152587.
[56] F. Rottensteiner et al., "The ISPRS benchmark on urban object classification and 3D building reconstruction," in *Proc. ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. I-3, 2012, pp. 293–298, doi: 10.5194/isprsannals-I-3-293-2012.

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